HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed, developed, and validated embedded and cloud-edge software modules for next-generation connected products, focusing on optimizing operational efficiency.
- Contributed to feature development, coding, unit testing, and software integration for specialised utility programmes, ensuring alignment with software specifications.
- Investigated, triaged, and resolved complex software issues across embedded and cloud-edge systems, reducing critical bug reports by **18%** over a 6-month period.
- Collaborated with cross-functional engineering teams, including systems, hardware, and test engineers, to deliver project objectives within Agile/Scrum frameworks.
- Conducted code reviews and regression testing to maintain high software quality standards, identifying and rectifying over **50 potential defects** prior to release.
- Produced comprehensive technical documentation for software projects, including design specifications and user guides, improving team onboarding speed by **30%**.
- Containerised microservices using **Docker** and deployed them to a cloud environment, managing CI/CD pipelines with **Jenkins** for automated testing and deployment.

SKILLS

Embedded Systems – Embedded Software Engineering, Secure Systems, Cloud-Edge Software, Software Design, Software Validation, Systems Engineering, Hardware Integration

Development – Software Development, Application Development, Utility Software, API Development, Unit Testing, Regression Testing, Debugging, Issue Triage

Databases – Database Management Software, SQL, PostgreSQL, MySQL, Oracle

Version Control – Git, Perforce, Source Code Management (SCM)

Cloud & CI/CD – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins

Languages – Java, Spring Boot, Python, SQL

PROJECTS

Embedded Security System for IoT Devices

- Developed secure communication modules for a fleet of embedded devices using **Java Spring Boot**, implementing mutual TLS for authenticated data exchange.
- Designed and integrated an API layer using **FastAPI** to manage device configurations and status updates from cloud-edge applications, achieving p95 latency below **200ms**.
- Wrote unit tests with **JUnit** and integration tests for the embedded components, ensuring **90% code coverage** for critical security functions.

Cloud-Edge Data Ingestion and Processing Pipeline

- Engineered a Python-based data pipeline on **AWS Lambda** to ingest and process telemetry data from connected devices, handling approximately **1,500 events per minute**.
- Implemented data validation and transformation logic using **pandas**, ensuring data integrity before storage in **S3** and upstream analysis.
- Containerised the pipeline components using **Docker** for consistent deployment across development and production environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)